



Project Name:	Trikon 14kW	Stage:	Sample C
Motor Code:	ML2A05C_Rev F	Created By:	GRV 08.01.2026
Revision:	V2	Approved By:	KSS 30.01.2026

CONFIDENTIAL

ML2A05C_Rev F Mechanical NVH Report

Initial Comments	Workbench Name - ML2A05C_Rev_F ITR 2 WB
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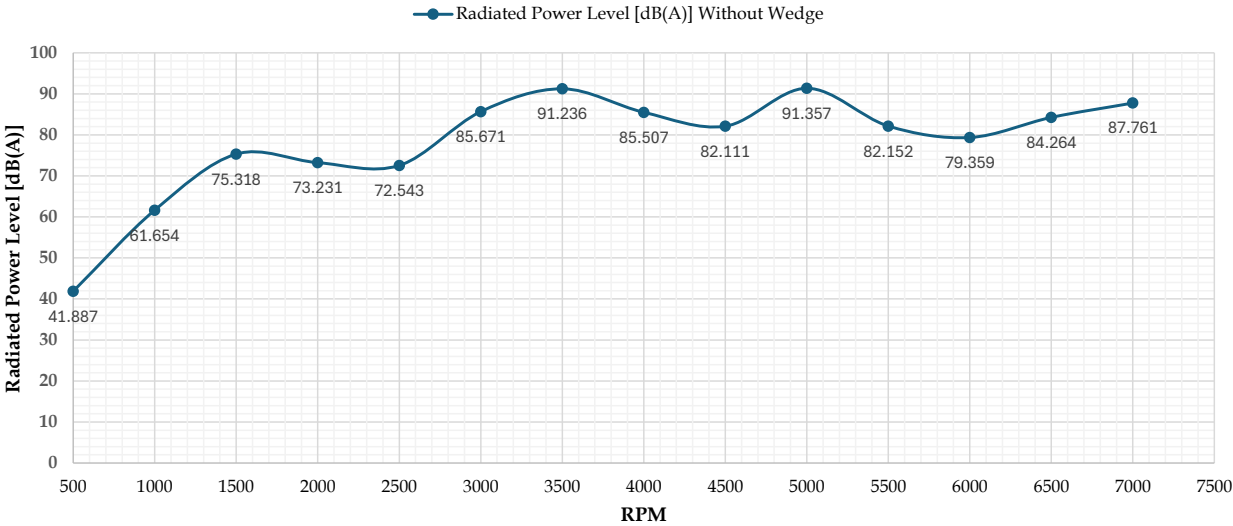
Static Structural Analysis	
Max Total Deformation	0.075 mm

Prestressed Modal Analysis	
Mode	Frequency (Hz)
2	825.83
3	851.02
4	996.40
5	1213.66
6	1244.22
7	1288.24
8	1454.31
9	1585.43
10	1623.54
11	1861.88
12	1954.51
13	1997.97
14	2095.69
15	2198.47

Harmonic Response Analysis		
Max Total Deformation	0.0024855 @5000	mm @ RPM
Max Total Velocity	38.69 @5000	mm/s @ RPM
Max Radiated Power Level	91.357 @ 5000	dB(A) @ RPM

Table - Peak Radiated Power Level All Harmonics [dBA]		On Outer Surface of the Housings	
Speed (RPM)	Radiated Power Level [dB(A)]	Frequency (Hz)	Order
500	41.887	-	Higher than 56
1000	61.654	-	Higher than 56
1500	75.318	-	Higher than 56
2000	73.231	-	Higher than 56
2500	72.543	-	Higher than 56
3000	85.671	-	Higher than 56
3500	91.236	1400	24
4000	85.507	1600	24
4500	82.111	1800	24
5000	91.357	2000	24
5500	82.152	2200	24
6000	79.359	2400	24
6500	84.264	-	Higher than 56
7000	87.761	-	Higher than 56

ML2A05C - Rev F Radiated Power Level [dBA]



Inference/Comments

Reference Images

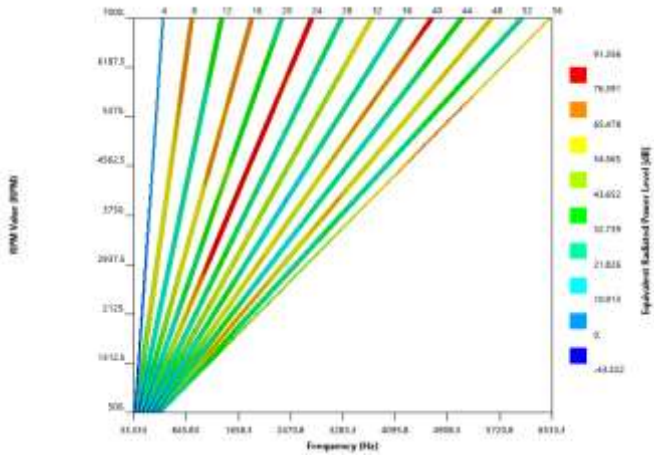


Fig. ERP Level Waterfall Diagram

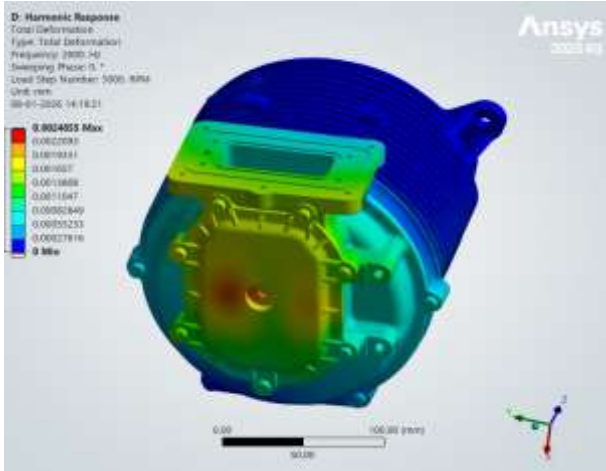


Fig. Max Total Deformation

Harmonic Acoustics Analysis

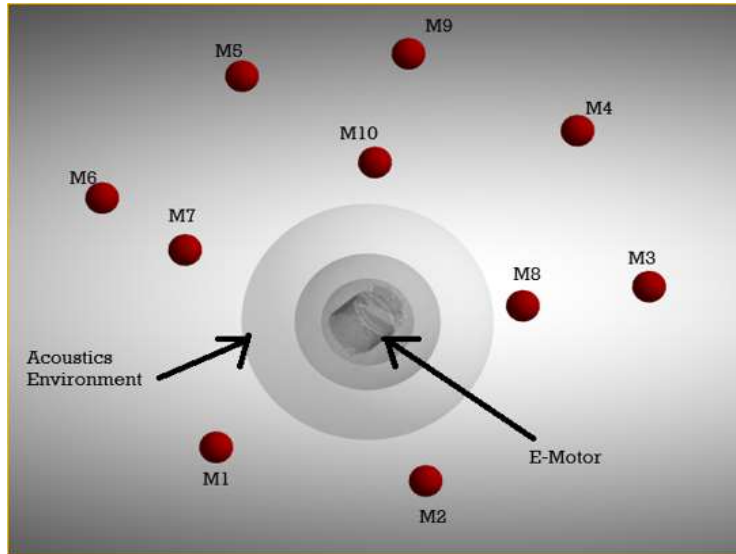
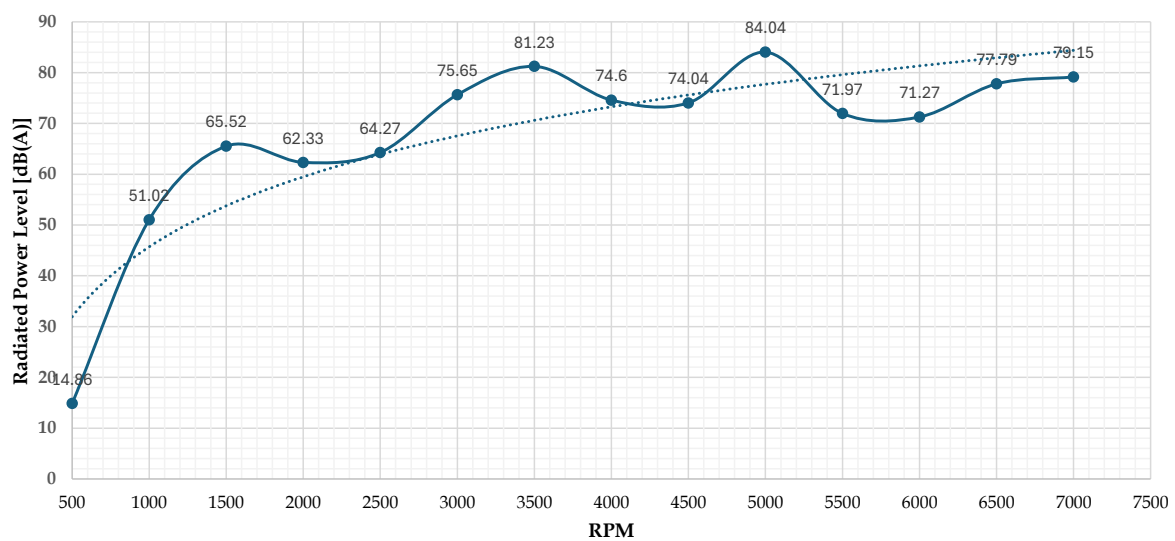


Fig. Microphone Positions

Table - Overall Radiated Power Level [dBA] 10 Mic Average at 1 Meter

Speed (RPM)	Radiated Power Level [dB(A)]
500	14.86
1000	51.02
1500	65.52
2000	62.33
2500	64.27
3000	75.65
3500	81.23
4000	74.6
4500	74.04
5000	84.04
5500	71.97
6000	71.27
6500	77.79
7000	79.15

Overall Radiated Power Level [dBA] 10 Mic Average at 1 Meter



Inference/Comments	1. The above acoustic results are based on simulation data and may vary under actual physical test conditions. These results will be refined once physical test data becomes available. 2. Peaks observed at certain operating speeds are attributed to resonance between electromagnetic (EM) force frequencies and the natural frequencies of the motor structure. 3. To mitigate these resonances, both EM and structural design modifications should be considered for the specific resonating frequencies.
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